

Irrigation and Water Management

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Preface

This book is primarily meant for undergraduate and postgraduate students of agriculture, horticulture, forestry, and agricultural engineering. It would also be useful for agronomists, irrigation engineers, agricultural engineers, ecologists, agricultural meteorologists, agricultural officers, and all others working in the field of water management. In short, the book has been moulded with two major objectives, to serve as a textbook for teaching “irrigation and drainage” or “water management” and to serve as a handbook for those involved in research and extension activities related to water management in agriculture. The stress is, however, directed towards small holder irrigation in the tropics.

Agricultural water management is an interdisciplinary subject wherein agronomists, irrigation engineers, agricultural meteorologists, and soil physicists interact. Those who work in water management sector must have a basic understanding of all these subjects, and this aspect has been considered well while formulating the outline of the book. Special attention has been given to update and make contemporary several concepts, which have undergone substantial changes in the recent decades in tune with international trends. I have tried to discuss the subject matter as simple as possible so that it is comprehensible to a broad spectrum of students and other stakeholders.

A course on water management or irrigation agronomy must provide a broad introduction to the edaphic, climatic, physiological, and environmental aspects of water balance in a farm unit. Adequate knowledge on these aspects is required to describe the soil-plant-water-atmosphere relationships logically. The first four chapters of the book have been devoted to describe these basic aspects. Efforts were also made to discuss some emerging topics like water foot print, virtual water, water security, and integrated water resources management.

The students and other stakeholders must be clear about the modern concepts of crop water use and irrigation water requirement. They must also be able to explain the theory behind irrigation scheduling and compute irrigation water needs of crops at the field level along with depth of irrigation and interval of irrigation. These aspects have been discussed in the chapters on “crop water use” and “scheduling irrigation to crops”. Various methods of irrigation are dealt at length in two chapters. More emphasis has been given to small-scale irrigation technologies such as micro irrigation as these technologies can improve the livelihoods of poor farmers in regions where water is scarce. A chapter has been devoted to describe various aspects of the quality of irrigation water along with relevant management options to cope with low quality water. The concept of water

productivity has been discussed in a separate chapter considering its worldwide importance. Water management recommendations for major crops of India have been provided in two chapters.

When the issue of drainage and problems of wet soils are discussed, one should also look at the status of wetlands, a much valued but often neglected ecosystem. A chapter has been exclusively set apart for discussing various aspects of wetlands in addition to a chapter on agricultural drainage.

Appropriate worked out examples have been provided wherever required. Illustrations include diagrammatic sketches and photographs for easy understanding of the subject matter. The units followed in the book are SI, but certain popular non-SI units are also included along with their SI equivalents. The bibliography comprises of literature cited as well as additional reading consulted mostly for preparing the narrative, the common knowledge part of the book.

While finalizing the manuscript, I have considered suggestions from students and colleagues as well. I should particularly mention some of my former graduate students— Dr Indulekha V.P, Dr Syama S. Menon, Dr Savitha Antony, and Dr Sindhu P.V (presently, all of them are Assistant Professors in KAU) who are very resourceful with ideas and suggestions. My thanks are also due to Dr P. Prameela and Dr Meera V. Menon, my former colleagues in the Department of Agronomy, for their assistance and cooperation. I also express my appreciation to Sri. Praveen K.P. for the elegant cover design. It is my pleasure to acknowledge the earnest efforts of Mr Jai Raj Kapoor, Mr Sunil Saxena, and all at Ane Books, New Delhi for bringing out the book neatly and beautifully.

I hope that the subject matter covered in the book is useful to all the stakeholders in agricultural water management, which may lead to improvement in both crop and water productivity.

C. George Thomas

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Water Resources Management

Water is an indispensable resource for the existence of life on earth. Green plants, the primary producers of food, require large quantities of water continually for their growth and sustenance. Timely water supply is an essential requirement for successful farming. In human affairs, water plays a wider role. If properly cared for, it is the harbinger of life. If neglected, it can be the cause of great disasters. Frequent occurrence of drought and shortage of water for irrigation and drinking are universal problems. When the drought is over, it is the turn of floods. Repeated destructive floods and flash floods destroy vast areas of land every year and take a heavy toll of human and animal lives. Managing the availability and use of water for productive and ecological purposes is a critical issue for sustainable development.

Rainwater is the primary source of water for all organisms. Plants utilize rainwater once it becomes part of soil as soil water. Animals can use it once it is stored in some storage structures such as ponds, tanks, lakes, wells, streams, or rivers.

1.1. IRRIGATION, DRAINAGE, AND WATER MANAGEMENT

Irrigation, drainage, and water management are frequently used terms in agriculture. Therefore, it is necessary to understand and distinguish these terms clearly.

Irrigation

If the rainfall in a region is in adequate amounts and well distributed, you can expect bumper crops, provided all other inputs are in order. Nevertheless, quite often, one has to confront two situations regarding the availability of rainwater. The first is a situation of too little rainwater as in arid and semi-arid regions. One has to face this situation during summer in humid and semi-humid regions as well. This situation can be tackled well by resorting to irrigation, the controlled application of water to soil for supplementing moisture indispensable for crop growth. The other is a situation of too much rainwater as in humid climates, where drainage

may become necessary. According to Brouwer and Heibloem (1986), a distinction can be made among these climatic situations warranting the requirement of water.

Humid climates: The regions with humid climates normally receive more than 120cm of rainfall per year, which is sufficient to cover the water needs of majority of crops, if well distributed. Surplus water may be the problem in such climates, and therefore, drainage rather than irrigation would be a necessity for successful crop production.

Sub-humid and semi-arid climates: In general, these regions receive rainfall ranging from 40cm to 120cm per year. The amount of rainfall is often not adequate to meet the entire crop water needs. Crop production during the rainy season may be possible, but during the dry season, it is feasible only with irrigation. This is typical of most North Indian states, where Kharif crops are grown depending on rainfall and Rabi crops with irrigation.

Semi-arid, arid, and desert climates: These regions receive rainfall in sparse amounts only, often less than 40cm per year. Normal crop production is impossible based on rainfall, and therefore, irrigation is indispensable.

The above pattern indicates that if the supply of water to crops through rainfall is insufficient, it must be supplemented through irrigation. Irrigation plays a major role in ensuring food security of nations by supplementing or replacing the need for natural rainfall. If there is no rainfall or when it is deficient, the entire water needs of crops is supplied through irrigation. This is what is meant by *complementary irrigation*. However, in a situation of some rainfall, but not enough to cover the water needs of crops, irrigation is planned in such a way that the rainfall and irrigation together meet the water needs of crops. This is *supplementary irrigation*. In short, farmers plan for irrigation under the following situations:

Inadequate rainfall: When the rainfall at a place is insufficient to meet the consumptive use requirements of crops, it is essential to apply water through irrigation.

Uneven distribution of rainfall: In certain cases, total rainfall in a region may be adequate but unevenly distributed over time and space. For tackling such unreliable temporal and spatial distribution of rainfall, irrigation becomes indispensable.

More number of crops over a year: Sometimes, the rainfall in a region may be adequate to grow one crop in a year, but with irrigation, more crops would be possible in rotation or in succession. For example, in many parts of India, Kharif crops are raised solely depending on rainfall, but for Rabi crops and summer crops (Zaid), irrigation is required.

In farmlands, irrigation has some other objectives too. Considering all these aspects, we can say that the following are the main objectives of irrigation.

1. When the rainfall at a place is insufficient for crop growth, irrigation is employed to supplement rainwater and capillary contribution from low water table.

2. Irrigation loosens the soil for proper plant root development.
3. Irrigation cools the soil and surroundings of plants, thus ensuring more favorable environment for plant growth.
4. Irrigation allows cropping throughout the year
5. Irrigations helps to reduce hazards from frost
6. Irrigation provides crops an insurance against short duration drought.
7. In certain cases, irrigation is done to dilute and leach down injurious salt contents present in the soil.
8. Pesticides and nutrients can also be applied through irrigation water (e.g., fertigation, herbigation, etc.).

Drainage

Along with the issue of dealing with water deficits, the situation of too much rainwater in a place must also be tackled well. Excess water problems can occur in all the climatic regions. If there is too much rainfall, soil get saturated with water, and a situation of *waterlogging* occurs. The process of removing surplus water from the soil surface or from the root zone of crops is called *drainage*. The requirement of drainage may be for a few days to ward off the problems from flash floods or throughout the year as in waterlogged areas.

Drainage facilities must be a part of farm management strategies, if the farm is prone to excess water. When we use the term ‘drainage’ in agriculture, it is *agricultural drainage*, the removal of excess water from farmlands. It is estimated that artificial drainage is being used on more than 10 percent of the farmlands in the world. Drainage is also needed for special requirements like leaching and reclamation of problem soils. Artificial drainage is a necessity in irrigated soils of arid and semi-arid regions to control salt concentrations in the root zone. Note that irrigation and drainage are considered complementary practices.

Water management

Control and management of water resources are important for their equitable and effective utilization. *Water management* envisages control of all human interventions on water resources. In a broader sense, every planning activity that has something to do with water can be looked upon as ‘water management’. It involves integrated management of surface water resources, ground water resources, and rainfall in an equitable way.

Water management is often defined as the process of integration of all the activities aimed at exploiting water in a technically and economically optimal way, but minimizing damages caused by it. *Agricultural water management* is water management in agricultural lands that includes both irrigation and drainage. *Urban water management* deals with the supply of drinking water, management of waste water, and tackling drainage problems in residential and other municipal areas. The process of agricultural water management includes the intake, conveyance, regulation, measurement, and application of water in appropriate quantities at the

right time to increase the production of crops. Agricultural water management also involve timely and effective drainage of excess water from the farms, which otherwise hamper crop production.

Management of water is an interdisciplinary subject, wherein agronomists, irrigation engineers, agricultural meteorologists, and soil physicists interact. Management of irrigation water with reference to crop production is the central theme of *Irrigation Agronomy*. At the same time, construction and maintenance of irrigation related structures such as reservoirs, wells, tanks, conveyance structures, and water lifting devices constitute *Irrigation Engineering*.

An irrigation project such as an irrigation dam or pond has a watershed area or more specifically a *catchment area* and a *command area*. The land area that can be irrigated by an irrigation project is called its command area as opposed to catchment area or watershed area from which the dam collects water drained through surface runoff. *Ayacut* is the Indian name for command area of a river and *Anicut*, the dam and the storage area. Command area can also be envisaged for a pond or well for the area irrigated through these sources.

In irrigated areas, we pursue *command area development* approach for its comprehensive development. In such areas, availability of water, the most critical factor in crop production is assured, and therefore, intensive farming of a specialized nature is practiced. The emphasis is to maximize production from unit area. In most states, Command Area Development Authority (CADA) has been formed for the overall development of command areas of irrigation projects. It is mainly though participatory irrigation management with the active co-operation of all stakeholders.

Unlike water management, *watershed management* is a different concept. In watershed management, all-inclusive management of the contributing area of surface runoff for a drainage entity such as a stream, river, or lake is considered. Remember that watershed is defined as a land area bounded by a natural ridgeline from which all the surface runoff drains to a common drainage point such as a stream, river, or lake. Watershed area approach is followed for the development of rainfed areas where availability of water is dependent solely on rainfall. An element of uncertainty and risk is prevalent in the production system as the rainfall may be erratic in amount, intensity, and distribution. Therefore, diversified and mixed farming systems are commonly practiced. Although the approaches are different, command area development should not be seen in isolation with watershed management. Command areas may face problems, if the watershed areas of the water storage structures such as dams are degraded resulting in silting and low storage.

Globally, sustainable use of water resources is a grave issue. The International Water Management Institute (IWMI) at Colombo, Sri Lanka under the aegis of the CGIAR (Consultative Group on International Agricultural Research) functions focusing on the sustainable use of water and land resources in developing countries. At the national level, the Indian Institute of Water Management (IIWM), Bhubaneshwar functions under the ICAR (Indian Agricultural Research Institute)

with the same purpose. The Water Technology Centre of the IARI (Indian Agricultural Research Institute), New Delhi is another important institute in India dealing with water resources and management in the country.

1.2. ROLE OF WATER IN SOIL AND PLANTS

Water has many ecological and physiological functions in the plant environment. Water constitutes 80-90 percent of the fresh mass of most herbaceous plants and over 50 percent of woody plants. However, water content is not uniform in all the plant parts. For example, the apical portions of roots and stems contain 90 percent or more of water. Leaves and young fruits are also rich in water, but as these organs mature, their water content decreases. Woody parts of trees may contain about 50-60 per cent water, whereas the stems of most cereals such as wheat contain about 60-70 percent, which at harvest time declines to 5-10 percent. Freshly harvested grains of most crops contain 10-15 percent of water. Moisture content of these grains determines their storage life, viability, and germination capacity. In general, crops require about 200 to 1000 kilograms of water for every kilogram of dry matter produced (Zonn, 1986).

Water acts as a reagent in many important physiological processes in plants such as photosynthesis. Water is essential to maintain the turgidity of leaves and new shoots and for the growth of cells. Remember that plant growth is affected by the rates of cell division and cell enlargement and by the availability of compounds essential for the synthesis of protoplasm and cell walls. A decrease in water content causes numerous problems in plants such as loss of turgidity, reduction in cell growth, cessation of cell enlargement, closure of stomata, reduction of photosynthesis, and interference with many basic metabolic processes. Continued dehydration causes disorganization of the protoplasm and finally death.

In photosynthesis, water is as important a reagent as carbon dioxide. A crucial function of water in plants is to act as a solvent through which gases, minerals, and other solutes enter plant cells and move from cell to cell and tissue to tissue within the plant. The permeability of cell walls and membranes to water helps to have continuous liquid phase in a plant through which translocation of solutes take place. All these indicate that water has several functions in plants. The role of water in plants can be summarized as shown below:

1. Water in plants creates turgor pressure providing turgidity and strength to plants.
2. Water is a reactant or substrate of many biochemical processes in photosynthesis and respiration.
3. Water acts as a solvent and carrier of many substances in plant cells and the medium in which biochemical reactions occur.
4. Water is necessary for the transpiration process in plants.
5. Water is important in stomatal opening. With water stress, stomata close and photosynthesis is affected.
6. Water is the transport medium for organic materials through phloem and inorganic nutrients through xylem.

7. Many organic constituents of plants such as carbohydrates, proteins, nucleic acids, and enzymes lose their physical and chemical properties in the absence of water.
8. Cell elongation or expansion is reduced or stopped with water stress. This indicates that water is essential for the growth of leaves, stems, roots, flowers, and fruits.
9. Water also acts as a temperature buffer and cools and hydrates plants.

In addition to the role water plays in plants, it has a crucial role in several soil processes as well. The following are the major roles water play in soil:

1. Water plays a pivotal role in soil formation by accelerating the process of weathering and soil profile development.
2. Water is the chief agent in soil erosion especially in the tropical regions.
3. Water affects many physical properties of soil such as structure, plasticity, aeration, strength, and swelling.
4. Water acts as a solvent, leaching agent, reactant, medium for chemical reactions, and plasticizing agent in the soil.

1.3. THE HYDROLOGIC CYCLE

The hydrologic cycle describes the continuous movement of water between the land, the ocean, and the atmosphere. The cycle has no beginning or end, but its many processes occur continuously in a recurrent manner. The most important processes of the hydrologic cycle are precipitation, infiltration, percolation, evaporation, transpiration, subsurface flow, groundwater flow, and surface runoff. It has many sub-cycles. All these are interrelated as could be seen from the typical hydrologic cycle portrayed in Fig.1.1.

Although it is true that there is neither a beginning nor an end to the hydrologic cycle, for convenience, we assume that the cycle begins from the oceans. Remember that 97.13 percent of water on earth is in the oceans. Because of solar radiation, oceanic water heats up, and the evaporated water escapes to the atmosphere. Evaporation also occurs from freshwater bodies, land, and vegetation surfaces. The evaporated water vapour is taken away through winds and lifted to the atmosphere finally forming rain-bearing clouds. Under favourable conditions, clouds condense and precipitate on the land and oceans as rain, hail, sleet, snow, or other forms. Before the precipitation reaches the earth's surface, some evaporation happens in the atmosphere. This increases atmospheric humidity, lowers temperature, and reduces evapotranspiration from crop plants. Water vapour may also be blown away by winds to the surrounding area. Although this fraction is partially useful, it is not considered in ordinary calculations because of the difficulty in its estimation.

When rainfall begins, a small part may be intercepted by the vegetation through the leaves and stems. Some of it may be absorbed and retained by leaves as *interception storage*; eventually lost as evaporation. If rain continues, the

storage on the leaves and stems become nearly constant, and all the water falling on them reaches the ground by drippings from the leaves or flowing down the stem. During light showers, the entire rainfall may be intercepted by the vegetation. The interception storage by vegetation may vary from 0.25mm to 9.14mm rainfall. For most crop plants, the mean value could be taken as 2.5mm.

The fate of rainwater falling on soil surface is not uniform. Some rainwater *infiltrates* into the soil; some may stagnate on the surface; some portion evaporates back to the atmosphere; while some may flow over the surface as run-off as a thin sheet of water and become the *overland flow* over the surface. A natural soil is usually protected by plant debris and vegetative cover. When this cover over the soil is ruined or destroyed, many problems can crop up. Foremost among them is a decrease in *infiltration capacity*. When infiltration is impaired, excessive overland flow occurs causing rapid run-off, often accompanied by increased erosion. Sometimes, percolation of water ceases because of the presence of a relatively impermeable layer of rock or clay pan in the subsoil. The infiltrating water then moves laterally as *interflow* or *subsurface flow* and joins the stream flow.

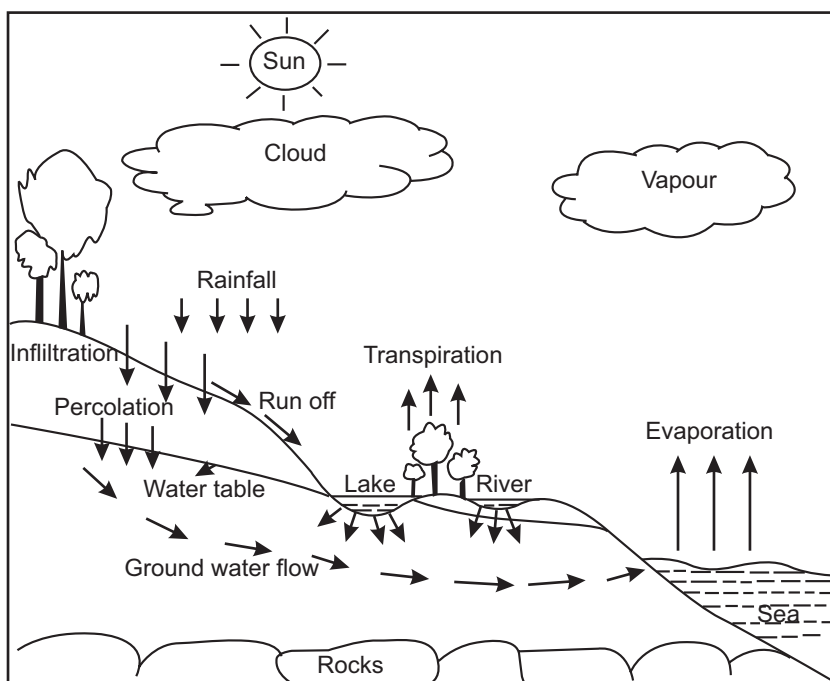


Fig. 1.1: The hydrologic cycle.

The rainwater stagnated on the soil surface can be considered as *depression storage*, which evaporates or infiltrate in due course. To a certain extent, it may be useful for meeting the water needs of crops, but may be harmful, if it persists on the surface for a long time, as it can create problems of water logging. Depression storage can be increased by various biological, engineering, or bioengineering conservation practices, which are helpful in reducing the threat of runoff. It has

been observed that contour furrows can store about 6.25cm of water and level bench terraces about 5cm.

After the entry of water into soil, some or all of it is held there by the molecular attraction of soil particles. Any additional water above this limit moves under the force of gravity to some surface outlets or aquifers. Water retained in the capillary soil pores against the force of gravity is called *capillary water* and this water is said to be in *retention storage*. The amount so held depends on the soil depth and other soil physical properties. This retained water will not move according to the forces of gravity, and therefore, not available for stream flow. However, it is available for plants; in fact, it is their principal source of water most of the time. Soil water under retention storage is prone to depletion by evaporation.

Water that infiltrates into the soil may be retained around the soil particles as a thin film in the root zone forming the *soil moisture*. When all the retention storage spaces of soil are fully occupied, the soil is said to be at *field capacity* (See Chapter 3). When this occurs, there would be still some storage spaces for moving water. In the absence of an impermeable layer in the subsoil, the infiltrating water percolates into the ground and builds up the *ground water* reservoir. The percolating water as ground water may move through the pores of saturated subsurface materials and may reappear at lower elevation as *underground flow* or *base flow* at the surface. Ground water coming out naturally at such places in the form of *springs* and *seeps* maintains the flow of streams and rivers, finally flowing out to the sea or evaporating into atmosphere as the hydrologic cycle continues. The water that moves as base flow is the source of all the perpetual flowing streams and recharging aquifers.

Large quantity of water is also stored in the form of *snow* on the polar areas and peaks of high mountains like Himalayas, which on melting feed the rivers originating from them. The streams and rivers carrying both surface and sub-surface flows eventually flow back to the seas and oceans. A large amount of the intercepted water and surface runoff returns to the atmosphere through evaporation. It is estimated that evaporation from the land surface consumes 61 percent of the precipitation and the remaining 39 percent flows as runoff to the seas and oceans. Evaporation from seas and oceans contribute nearly 90 percent of atmospheric moisture. Sometimes, water may also directly move to the atmosphere by sublimation and vaporization of the solid or frozen phase bypassing the liquid phase.

In a watershed, evaporation, transpiration, and runoff—the three major phases of the hydrologic cycle—can be altered, modified, controlled, or regulated by vegetation, better land husbandry, and structural management practices. There are several proven techniques, which control and regulate evapotranspiration and runoff.

1.4. PRECIPITATION: THE PRIMARY SOURCE OF WATER

Precipitation is the primary source of water on earth, and rainfall is the chief form of precipitation. *Precipitation* is a broader term that includes any sort of liquid

or solid form of water falling on earth's surface. In India, the monsoons (south west monsoon and north east monsoon), cyclonic depressions, and local storms are responsible for rainfall in varying degrees. The precipitation of water drops from clouds with more than 0.5 mm diameter is called *rainfall*. However, typical raindrops are with a diameter of 1-2 mm. In the tropics, much larger raindrops are received mostly in the order of 5 mm in diameter. *Shower* is the name given to a rainfall event that ceases as suddenly as they start, but the size of raindrops is larger and could be up to 8 mm. A *drizzle* is precipitation having fine drops smaller than 0.5mm diameter, and because of very small droplets, drizzle is not considered as rainfall. *Virga* is the name for raindrops that evaporate before reaching the ground. *Freezing rain* includes raindrops having temperatures less than 0°C, which freeze upon contact.

Precipitation of small ice crystals falling on earth's surface is called *snow*. Snow fall takes place in extreme cold situations, where the ice crystals would not get enough time to melt before reaching the ground. Snowfall is a common phenomenon in colder countries and mountainous regions such as the Himalayas. The name *sleet* is given to the combined precipitation of rainfall and snow, which involves the mixing of solid and liquid forms of water. Sleet is generally composed of ice pellets, frozen raindrops, and partly melted snowflakes. Probability of sleet is high when air temperature at the surface approaches 1.5°C. The precipitation of aggregates of icy matter or frozen water drops is called *rime*. *Graupel*, also called *soft hail*, is particles of snow with a fragile crust of ice. The loosely compacted ice crystals of graupel, often less than 5mm diameter, are almost spherical and breaks upon striking the ground. The phenomenon of falling bigger lumps of ice is called *hail* or *hailstones*.

Please note that localized phenomena such as *haze*, *mist*, *fog*, *smog*, *dew*, and *frost* do not come under the purview of precipitation. These occur when there is a clear sky, weak radiation, and no winds. *Haze* occurs because of condensation of water vapour on microscopic condensation nuclei such as fine dust, salt particles, and soot particles. When the size of these particles is very small (about 0.1 μm), it is called *dry haze*. As relative humidity increases, dry particles may become wet. Damp haze blocks more light than dry haze. Both *mist* and *fog* occur from condensation of water vapour in the air on the surface of either land or water. However, mist is differentiated from fog due to better and longer distance visibility. *Smog* denotes *smoke* + *fog*, a coined term to describe the greyish fog formed in and around industrial townships.

Dew is condensation of water vapour on solid surfaces that are cooled due to heat loss by night radiation, and temperature is brought below the dew point. Dew formation takes place when air cools to the dew point, so that water vapour condenses on various surfaces such as grasses and vehicles. Dew may be beneficial to crops in moisture stressed areas. Frozen dew as spherical beads forms when the temperature drops below the freezing point. *Frost* occurs when water vapour is deposited on a surface, which is below the freezing point.

Characteristics of rainfall

Most crops meet their major share of water from rainwater, the primary source of soil moisture. Several characteristics of rainfall affect the fate of rainfall. The *amount*, *duration*, *intensity*, *frequency*, *distribution pattern*, and the *time of the season* are important characteristics of rainfall, which have a bearing on plant growth and the environment.

The *amount of rainfall* means total rainfall received during a particular event or period, and the *duration* means the length of rainfall received in a particular rainfall event or total rainfall period. The *frequency* denotes how frequent the rainfall is in a particular region. The *distribution pattern* indicates how the rainfall occurs in various parts of the year or for how many days a region receives rainfall. It is usually indicated by rainy days and the month wise spread. The *intensity of rainfall* refers to quantity of rainfall in relation to time of its occurrence. In other words, intensity is the amount of rainfall per unit time. Please note that only those days that received a total rainfall of 2.5 mm or above are considered as *rainy days*.

Rainfall is measured by means of *rain gauges*, and is expressed in terms of depth to which rainwater stands on an area, if all the rainfall is collected on it. For example, 10 mm rainfall means a downpour at the rate of 10 L per square metre area (10L/m^2). Rain gauges can be non-recording and recording. The non-recording types consist of a cylindrical vessel assembly kept in the open (Fig. 1.2). The collected rainfall is measured with the help of a graduated cylinder. Rainfall for 24 hours is reported as depth of rainfall by dividing the volume of water collected with the aperture area of the gauge in cm or mm. Recording type rain gauges do not measure total rainfall but records it by a special arrangement (Fig. 1.3). Using this type of rain gauge, one can observe the onset, the cessation, duration, and intensity of rainfall. The data are useful for soil erosion and stream flow measurement studies.



Fig. 1.2: Standard rain gauge to measure the amount of rainfall.

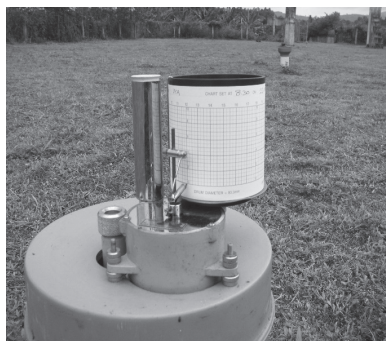


Fig. 1.3: Self-recording rain gauge.

Rainfall varies both in time and space. The variability can be visualized by analyzing the rainfall records of different gauging stations. In India, the annual rainfall varies from about 1100 cm in the northeastern part of India to as low as about 20 cm in the north western parts with an average value of 119 cm. Understand that rainfall distribution is more important for crops than total amount or intensity. Although it is difficult to predict rainfall characteristics for a given season in an area, one can utilize mean values based on long-term records for determining the kind of water management measures required for different crops.

Monsoons in India

Rainfall in India is largely due to monsoons. The monsoons are seasonal winds that develop because of changes in atmospheric pressure caused by varied heating and cooling of continental landmasses and oceans. The success of agriculture in India depends greatly on the timely occurrence, receipt of adequate amount, and even distribution of rains during the crop seasons. Remember the much cliched saying, “agriculture in India is a gamble with the monsoons.” Although we have come a long way after bringing more area under irrigation, still, the importance of monsoons holds good. The amount and distribution of rainfall received during the months of July and August are critical as the fate of rainfed Kharif crop is highly dependent on rainfall during these months. The failure or break in monsoons could spell disaster not only to agriculture but also to all the economic activities.

The most well-known monsoons in the world are those affecting India and Southeast Asia. In India, two monsoons occur based on the region of origin of monsoon winds—the *southwest monsoon* and the *northeast monsoon*. The summer monsoon (southwest monsoon), which blows southwesterly across the Indian Ocean is extremely wet. In contrast, the winter monsoon, which blows north easterly (northeast monsoon) is generally dry. Apart from affecting India and Southeast Asia, the monsoons have some influence on the climate of Central Africa and northern Australia.

During summer months, the sun moves northwards to the Tropic of Cancer, and large land masses of Central Asia and the Indian subcontinent get heated up generating a seasonal continental region of low pressure. Some areas of northern and central India reach temperatures of more than 40°C. The hot air that rises over Southeast Asia during April and May creates low-pressure areas. The Indian Ocean to the south of India is not heated up to the same extent, and therefore, serves as a centre of high pressure. When the wind blows south-westerly across the Indian Ocean, it accumulates considerable moisture. Clouds are formed, from which heavy rainfall occurs during the wet season from May to September.

The great Himalayas act as a wall causing the warm air to drop its moisture over southern Asia. The southern sides of the Himalayas receive good amounts of precipitation, but the northern sides receive small amounts only. This orographic effect is also seen along the southwestern coast of India because of the Western Ghats. Consequently, certain areas of India receive considerable amounts of precipitation during the summer monsoon season. However, regions on the

leeward side of the Western Ghats like Coimbatore in Tamil Nadu get very less rains during the monsoon season.

The South West monsoon originates in the south of equator, moves forward and touches India over the Andaman and Nicobar Islands by 20 May. In normal situations, the southwest monsoon reaches the Kerala coast by 1 June. It then moves along the Konkan coast in early June, and covers the entire country by the end of July. The rains continue up to the end of September and then the southwest monsoon recedes. In November-December, the northeast monsoon is the main contributor to the rainfall over the southern portion of the peninsula.

In fact, the southwest monsoon advances as two branches because of the features of the Indian peninsula and the western ghats. The branch known as the *Arabian Sea monsoon* after breaking apart at the southern part of the peninsula reaches Mumbai around June 10. By late June, this branch spreads over the entire South Asia fetching cooler and humid weather. The other branch, often called the *Bay of Bengal monsoon*, advances northwards in the Bay of Bengal region and covers Assam by the first week of June. Because of the barrier effect of the Great Himalayan Ranges, this monsoon is deflected westwards along the Indo-Gangetic Plains towards New Delhi. By July, the two branches of monsoon merge causing precipitation along other parts of North India. Note that these are the normal dates of onset of monsoon, but it may be delayed or may even be early in some years.

Retreat of Monsoons : After giving sufficient rains all over India, the monsoon starts withdrawal from northwest India by the beginning of October and from the remaining parts of the country by the end of November. These retreating winds lead to the formation of northeast monsoon over the southern half of the peninsula. As northeast monsoon is because of this retreat of winds, it is also called the 'retreating monsoon'. This is a classic example of change of direction of winds consequent to the migration of the sun southwards heralding the beginning of winter in the Northern hemisphere. The Indian subcontinent cools fast but the Indian Ocean remains almost warm. The warm waters heat up air masses over the oceans, driving them to move upwards. This upward movement of warm winds drive the cool air from the Himalayas to the Indian Ocean. Although northeast monsoon is a dry wind system, because of the development of cyclones at the head of the Bay of Bengal, which cross over peninsular India, the winds pick up moisture from the Bay of Bengal causing precipitation in the east coast. The Indian states of Tamil Nadu, Karnataka, and Kerala receive good amount of rainfall from the northeast monsoon from Nov.to Dec.

Pre-Monsoon Showers : There are several places in India, which receive good summer showers or pre-monsoon showers. The months from March to May are considered as the period of pre monsoon rains. However, it is difficult to predict their actual time of arrival or occurrence. Pre monsoon showers are due to convective precipitation. It is of very short duration and the intensity may range from light showers to heavy thunderstorms along with forceful winds. Sometimes, it also causes hailstorms. Heating during daytime before noon triggers the convection process resulting in thunderstorms in the afternoon. Often, a day may

begin with a clear sky and hot weather, but in the afternoon, showers start pouring, and the accompanying rain cooled winds cause a reduction in temperatures.

Pre-monsoon showers are important for Kerala, Karnataka, Konkan, and Goa. It is called 'mango shower' in these areas because it prevents the mango from falling from the trees before ripe and also helps in ripening. In some areas of Kerala (Wayanad) and Karnataka, it is popularly known as 'cherry blossom showers' or 'coffee showers' because it triggers blossoming of coffee. In Kerala, pre-monsoon rains proclaim the starting of farming activity after a brief slump due to summer. Planting of crops like cassava, yams, taro, ginger, turmeric, and the first crop of rice (Virippu) begins immediately after the receipt of pre-monsoon showers. The summer rains are important in tea growing areas, and in Assam, it is called 'tea showers', as these rains increase the quality and productivity of tea.

In the Gangetic plains of India, thunderstorms locally known as 'Kal Baisakhi' or 'Nor'westers' bring hot weather rainfall between April to May. It is called Kal Baisakhi as they occur in the Baisakhi month. Nor'westers bring good amount of rainfall in the afternoon or evening hours of pre-monsoon months over Bengal, Bihar, Jharkhand, Chhattisgarh, Assam and adjoining areas. Sometimes, these are destructive in terms of lightning, thunderstorm, hailstorm, and torrential rainfall. Nonetheless, it is helpful for the pre-Kharif crops such as jute, rice, and several fruits and vegetables.

Winter Rains : Precipitation can also occur during the cold season in North and Central India. It is because of a low-pressure system in North India called western disturbance. This advances from west to east and brings snow fall in the Himalayas and rainfall in the submountain and nearby areas. The precipitation, however, reduces towards the east. Rainfed Rabi crops of North and Central India depends heavily on this winter rainfall.

Rainfall pattern in India

The mean annual rainfall over India is 119cm. On an average, 76 per cent of the total rainfall is received during the southwest monsoon season (June-Sept.), and 10 percent each is received during the post monsoon season (Oct-Nov.) and summer season (March-May). The remaining 4.0 per cent is received during winter months (Dec.- Feb.). This is the normal trend in most North Indian states. However, in the south and southeast regions, substantial rainfall occurs during Oct. to Dec. under the influences of northeast monsoon. For example, Tamil Nadu receives 49 per cent of the total rainfall during Oct. to Nov. from northeast monsoon. In the country, there are regions, which receives rainfall in the range of about 1100 cm (Northeast regions), but in Rajasthan, it is very meagre, often less than 10 cm. As most rainfall is confined to 3 to 4 months of the year, assured water supply to agriculture, industries, and drinking purposes is a great challenge. Note that except Jammu and Kashmir in the extreme north and Tamil Nadu in the south, 80-90 percent of the total rainfall over the country occurs during the southwest monsoon period from June to Sept.

1.5. RAINFED AGRICULTURE AND IRRIGATED AGRICULTURE

Land areas of the world are often classified into *arid*, *semiarid*, and *humid* regions based on precipitation and evapotranspiration. A most widely used yardstick for such a classification is that of Thornthwaite and Mather (1955). In this system, *moisture deficit index* is calculated considering precipitation (P) and potential evapotranspiration (PET), and based on the index, climatic types are identified. It gives an idea about the rate of utilization of soil moisture for evapotranspiration when precipitation (P) falls short of potential evapotranspiration (PE).

$$\text{Moisture Deficit Index (MDI)} = \frac{100 (P - \text{PET})}{\text{PET}}$$

Where, P-Precipitation and PE-Potential evapotranspiration.

This system of classification is followed in India to demarcate arid, semiarid, and humid zones. Accordingly, Saurashtra, Kutch, and West Rajasthan are the major arid regions in India. Bellary-Anantpur area in Karnataka, Andhra Pradesh and Telangana states and Tirunelveli region of Tamil Nadu also come under arid zone. Almost the whole of peninsular India east of Western Ghats, except for its north-east portion and small areas around Visakapatanam-Kalingapuram area in coastal Andhra Pradesh and Telengana and Gaya-Jamui area in Bihar fall under semi-arid zone. The West coast and adjoining Ghats, the Himalayan States, the Northeastern states, small areas at high altitude in Aravalli and Vindhays, and the Islands come under humid to per-humid zone. The remaining areas of India are classified as sub-humid.

Table 1.1: Moisture regions based on the climatic classification of Thornthwaite and Mather (1955)

<i>Climatic type</i>	<i>Moisture index</i>
A– Per-humid	100 and above
B ₄ –Humid	80 to 100
B ₃ –Humid	60 to 80
B ₂ –Humid	40 to 60
B ₁ –Humid	20 to 40
C ₂ –Moist sub-humid	0 to 20
C ₁ –Dry sub-humid	–33.3 to 0
D – Semi-arid	– 66.7 to –33.3
E –Arid	–100 to – 66.7

Dryland Agriculture

Crops grown in different climatic situations vary greatly with respect to the requirement of water. In certain areas, irrigation is not required, rainfall is sufficient to meet the water needs of crops. Crop production that is entirely dependent on

rainfall is called *dry land agriculture*. However, dependability of crop production varies based on the quantum and distribution of rainfall. Dry land agriculture is often grouped into three categories—*dry farming*, *dry land farming*, and *rainfed farming* based on the quantum of rainfall and the risks involved (Reddy and Reddi, 1992). Although arbitrary and the boundaries are thin, this classification is most often used to distinguish the requirements of water for different farming situations.

Dry farming: The practice of crop production without irrigation in dry areas that receives a rainfall of 75 cm or less annually is often called dry farming. In most of the arid areas, dry farming is common. The term “dry farming” is a misnomer; it is simply the farming system followed under drier conditions using improved water conservation technologies. Dry farming always implies farming under a comparatively small annual rainfall. Crop failures are more common under these conditions. In such areas, crop production even during the rainy season is unreliable. Crop production in dry farming tracts heavily depends upon the timely receipt of meagre rainfall and efficient moisture conservation practices. Otherwise, irrigation must be an integral part of the crop production strategy.

Dry land farming: Crop production without irrigation in areas where the annual rainfall is in the range of 75 cm to 115 cm per year is usually called dry land farming. Dryland farming is common in the semi-arid and sub-humid areas. The amount of rainfall is not sufficient to cover the water needs of the crops. In such regions, especially in the dry season, crop production is possible only with irrigation. In the rainy season, however, crop production may be possible without irrigation by utilizing rainfall. Dry spells during the crop period may occur but crop failures are rare during the rainy season. Often, the growing period extends from 75 to 120 days.

Rainfed farming: Crop production in humid and per-humid regions receiving more than 115 cm rainfall is sometimes called rainfed farming. The rainfall is often adequate to cover the water needs of various crops. Crop failures are rare. Tree crops are predominant in such regions. However, drainage may become necessary to dispose excess water. The growing period would be more than 120 days.

Irrigated Agriculture

Dry farming or dry land farming techniques cannot ensure the success of crop production simply based on the meagre rainfall and conserved water. Although in the rainy season, crop production may be possible without irrigation by utilizing rainfall, this is not enough to ensure continuity of food production for the masses. In such regions, irrigation must be an integral part of the crop production system. When the crop production is done dependent on irrigation, it is called *irrigated agriculture*.

The total geographical area of India is 329 million ha. Net cultivated area remains almost at 141 million ha for the last 40 years. Net irrigated area in the country for the past few years is hovering around 68 million ha (48%) and rainfed area 73 million ha (52%) (GOI, 2020).

1.6. WORLD WATER AVAILABILITY AND WITHDRAWAL

Water covers nearly 70.8 percent of the globe. However, most water on earth, a staggering 97.47 percent is saltwater that fills the oceans (1351million km³). Seawater contains about 3.5 percent of dissolved salts, predominantly sodium chloride. Only the remaining 2.53 percent (35 million km³) is fresh water. Out of this fresh water, 69.7 percent is present in polar and high-altitude regions as ice caps, icebergs, and glaciers, which is unavailable for human use. About 30.0 percent fresh water occurs as ground water, of which about half the volume lies in water bearing formation deeper than 800m below ground surface and is not ordinarily available for economic use. About 0.3 percent fresh water (0.1 million km³) is available as surface water in lakes and rivers and as atmospheric moisture. The above assessment indicates that the amount of water available for use is mainly that in the rivers and its numerous streams, freshwater lakes, and about half of the ground water reserves.

Mean annual rainfall over the world is estimated to be 99cm. On volume basis, this amounts to 119,000 km³ over the land, of which about 74,000 km³ (62%) evaporates back into the atmosphere (FAO, 2002). The remaining 45,000 km³ (38%) flow into lakes, reservoirs, and streams or infiltrate into the ground to replenish the aquifers. Although this represents the 'renewable water resources', not all these 45,000 km³ are accessible for use because of many reasons. A part of this rainwater flows into distant rivers, loses during seasonal floods, or percolates deeper into groundwater. An estimated 9,000km³–14,000km³ can be considered economically available for human use.

Annual withdrawal of water for human use is estimated to be about 3600 km³, the equivalent of 580 m³ per capita per year (FAO, 2002). All the surface water should not be utilized for human use, a part of it must be allowed to follow its natural course for the conservation of the aquatic ecosystem. The instream flow requirement is estimated at 2350 km³ per year. This amount and the amount withdrawn for humans together come to 5950 km³. This can be considered as the easily accessible freshwater resources. Human water use has increased more than 35 folds over the past three centuries. As both water and population are unevenly distributed and some countries are overpopulated or water scarce, the situation of global water supply is already grim. Many regions of the world are suffering from freshwater shortages, and competition among the users for water is rising unabated.

Agriculture is the biggest user of water, accounting for about 69 percent of all withdrawals worldwide. Domestic and municipal use amounts to about 10 percent and industry uses some 21 percent. Water use varies considerably around the world. Most of the developed countries use less water for agriculture and more for industrial and domestic uses. For example, in India, the figures are 84 for agriculture, 12 for industry and 4 for domestic uses. In Kerala, 71 percent is utilized for agriculture, 11 percent for industry and 18 percent for domestic uses.

Water Availability in India

Based on the mean annual rainfall of 119.4 cm over a geographical area of 328 million ha, total annual rainfall is estimated at 392 M ha-m (Michael, 1978). Accounting the contribution from snow fall, this can be rounded off to 400 M ha-m. Out of the total annual precipitation of 400 M ha-m, about 70 M ha-m is lost to atmosphere as evaporation. The remaining is 330 M ha-m, and out of it, about 115 M ha-m flows as surface runoff and 215 M ha-m infiltrates into the ground. About 77 percent of the infiltrated water (165 M ha-m) is retained as soil moisture and about 23 percent (50 M ha-m) percolates deeper and joins the water table. Note that this 50M ha-m, which percolates deeper into the ground, comes to about 12.5 per cent of the total precipitation of 400M ha-m. Along with this, there occurs some influent discharge from flood flows (5M ha-m) and from irrigation systems (12 M ha-m). Thus, 67 M ha-m of water may be present as ground water (Michael, 1978).

In 1997, the Central Ground Water Board of India in collaboration with the respective State Departments assessed the ground water resources of India at 43.0 M ha-m (Chadha, 1999). The availability of renewable ground water resource for irrigation has been computed at 36.09 M ha-m.

1.7. WATER RESOURCES FOR IRRIGATION

India, which has more than 17 percent of the world's population, has only 2.45 percent of world's land resources and 4.0 percent of the world's freshwater resources. As discussed earlier, monsoon rains are the main source of fresh water in India, and rainfall is dependent on the southwest monsoon, northeast monsoon, shallow cyclonic depressions and disturbances, and local storms. Water for irrigation is obtained from:

1. Surface water resources such as rivers, streams, ponds, tanks, lakes, and artificial reservoirs like dams, barrages, and weirs, and
2. Ground water resources such as springs, dug wells, and tube wells.

Surface Water Resources

Surface water resources include the natural flowing water through the rivers and streams and the still water stored in ponds, tanks, lakes, or artificial reservoirs such as dams, barrages, and diversionary weirs. Precipitation including rainfall and melting snow act as the primary contributor to various surface water resources.

Rivers

Rivers and their tributaries and stream network constitute the major share of surfacer water resources. A river is a big natural water course flowing towards an ocean, a sea, an estuary, a lake, or another river. India is blessed with numerous rivers spread over the entire country. These rivers can be broadly classified into two groups, the *perennial rivers of the Himalayan region* and the *rainfed rivers of peninsular India*.

The great rivers of North India, which are fed by the melting snows and glaciers of Himalayan region during summer and overland flow and subsurface flow during the monsoons are perennial but often unpredictable. The lean period of these rivers is the winter, but at no time of the year, the flow is reduced as in the case of peninsular rivers. The main Himalayan river systems are those of the Indus, the Ganges, and the Brahmaputra along with their tributaries. The Indus River rises north of Mansarovar in Tibet and flows through Kashmir for a distance of 650 km and enters Pakistan. The tributaries in the plains are the Jhelum, the Chenab, the Ravi, the Beas, and the Sutlej. The Chenab flows through Himachal Pradesh. The Ravi and the Beas flow through Punjab and Sutlej forms the boundary between India and Pakistan. The Ganges rises in Gangotri and travels 2525 km before joining the Bay of Bengal. The most important rivers feeding the Ganges are the Yamuna, the Ghaghara, and the Kosi. The Ganges has the largest catchment area (86.1 M ha) in India.

The Brahmaputra river originates from the Kailash ranges of Himalayan mountains at an elevation of 5300m. After flowing through Tibet, it enters India through Arunachal Pradesh and flows through Assam and finally through Bangladesh before it joins Bay of Bengal. The river has the distinction of passing through four countries—China (Tibet), India, Bhutan, and Bangladesh.

The rainfed rivers of peninsular India fall into two categories, the *coastal rivers* and *inland rivers*. The coastal rivers are comparatively small streams. About 14 percent of the water resources of India are in these rivers. The inland rivers are of great antiquity. Those flowing westwards are the Narmada and the Tapti. The east flowing rivers are the Mahanadi, The Brahmani, the Godavari, the Krishna, and the Kavery.

Estimates on surface water resources: For estimating the surface water resources of India, Dr. A. N. Khosla divided the country into six water resources regions (Khosla, 1949). Khosla estimated the mean annual runoff from the six water resources regions of the country at 167.23 M ha-m (Table 1.2).

Table 1.2: Water resources regions in India (Khosla, 1949)*

<i>Name of basin</i>	<i>Catchment area (M ha)</i>	<i>Annual runoff (M ha-m)</i>
Indus system*	35.40	7.94
Ganga system	97.60	48.96
Brahmaputra system*	50.62	38.08
Rivers falling into Arabian sea excluding Indus	49.16	41.19
Rivers falling into Bay of Bengal other than Ganga and Brahmaputra system	121.03	31.06
Rajputana (Luni and Ghaggar basins)	16.80	-
Total	370.61	167.23

*The catchment areas of the Indus and Brahmaputra basins extend beyond the boundary of India.

According to the estimates provided by the Central Water Commission (CWC, 2020), annual average flow through major Indian rivers is 1869 km³ and the utilizable annual surface water of the country is 690 km³ (Table 1.3). Utilizable water resource means the amount of water, which can be withdrawn from its place of natural occurrence.

Table 1.3: Water resources potential in the major river basins of India (CWC, 2020)

<i>Sl. No.</i>	<i>River Basins</i>	<i>Average Annual Flow (km³/year)</i>	<i>Utilizable Flow (km³/year)</i>
1	Indus (up to border)	73.31	46.00
2	Ganga-Brahmaputra Basin		
	a. Ganga	525.02	250.00
	b. Brahmaputra, Barak, and others	585.60	24.00
3	Godavari	110.54	76.30
4	Krishna	69.81	58.00
5	Cauvery	21.36	19.00
6	Pennar	6.32	6.32
7	East flowing rivers between Mahanadi and Pennar	22.51	13.11
8	East flowing rivers between Pennar and Kanyakumari	16.46	16.73
9	Mahanadi	66.88	49.99
10	Brahmani & Baitarni	28.48	18.30
11	Subarnarekha	13.37	6.81
12	Sabarmati	3.81	1.93
13	Mahi	11.02	3.10
14	West flowing rivers of Kutch, Saurashtra including Luni	15.10	14.98
15	Narmada	45.64	34.50
16	Tapi	14.88	14.50
17	West flowing rivers from Tapi to Tadri	87.41	11.94
18	West flowing rivers from Tadri to Kanyakumari	113.53	24.27
19.	Areas of island drainage in Rajasthan desert	Neg.	-
20	Minor river basins drainage to Bangladesh & Myanmar	31.00	NA
	Total	1869.35	690.32

Irrigation commission of India (GOI, 1972), based on the specific analysis of the stream flow data from 80 sources, has worked out the water resources of the

country at 178 M ha-m. This includes, in addition to surface run off, groundwater runoff contributed by the ground water storage as base flow of the rivers as well as the subsurface runoff contributed by rainfall and seepage from surface water resources (45M ha-m). It also includes about 20M ha-m brought in by streams and rivers from catchments lying outside the country. The remaining 115 M ha-m is direct contribution by precipitation inclusive of snowfall amounting to about 10 M ha-m. The fate of surface water flow is as shown below.

- ◆ Stores in reservoirs
- ◆ Utilizes directly by diversion or pumping
- ◆ Loses as deep percolation from streams and rivers
- ◆ Flows to seas and oceans
- ◆ Loses due to evaporation
- ◆ Loses as seepage

Reservoirs

A reservoir is often used to store water, which may be an enlarged lake (natural or artificial), storage pond, or impoundment from a dam. Reservoirs are created in many ways including controlling a river, or interrupting a watercourse by building an embankment, or excavated in the ground. Reservoirs are often created by dams, which not only suppress floods but also provide water for many human activities. A barrier constructed across a river to stop or restricts the flow of water is called a *dam*. Often, dams are constructed in a valley considering the natural topography to reduce the construction costs and to store maximum flow. In India, water flowing through various rivers and streams can be used for irrigation purpose directly or by constructing reservoirs through dams and taking water through canals.

Classification of Irrigation Projects

The irrigation schemes constructed in India are classified as major, medium, and minor irrigation projects depending upon the culturable command area (CCA) envisaged. In this connection, one has to familiarize with three terms, command area (CA), gross command area (GCA), and culturable command area (CCA).

Command area is the area that can be irrigated by a source of water or canal system. *Gross command area* (GCA) is the total theoretical area that can be irrigated by a source of water or canal system on the perception that unlimited quantity of water is available. However, this may include residential areas, roads, townships, storage ponds, and uncultivable areas, which would not be irrigated.



Fig. 1.4: Vented cross bar (VCB), a minor irrigation structure.

Culturable command area is the area actually irrigated within the GCA. However, if the required quantity of water, fertilizer, and other inputs are not available at a particular point of time, the entire CCA may not be put under cultivation during any crop season. Similarly, the land may be kept fallow purposefully for one or more crop seasons to increase the fertility of the soil. In some areas of the CCA, because of high water table, irrigation need not be given as the crops get enough water from capillary rise. Thus, during any crop season, only a part of the CCA is put under cultivation, and this area is called the *culturable cultivated area*. Naturally, the remaining area which is not cultivated during a crop season is the *culturable uncultivated area*. In India, irrigation projects are classified mainly based on CCA.

Major Project: A major irrigation project comprises of huge surface water storage reservoirs and flow diversion structures. A major project usually has a culturable command area of more than 10,000 ha.

Medium Project: Surface water irrigation projects having medium size storage and diversion structures are called medium projects. The CCA of such medium project extends from 2000ha to 10000 ha.

Minor Project: The CCA of minor irrigation projects is below 2000ha. The project is usually based on springs, wells, tube wells, surface water lifted by pumps, or tanks. Vented cross bars (VCB) erected in streams to divert surface flow also comes under minor irrigation project.

Canal Irrigation

Water flowing through rivers, streams, and other surface water resources can be used for irrigation directly or by diverting through canals from big reservoirs constructed for the purpose. In most cases, dams and weirs are constructed across the rivers, and water is diverted through canals for irrigation. In India, canal irrigation accounts for about 30 per cent of the net irrigated area. Canal irrigation is possible in areas that are drained by well-distributed perennial rivers. After independence, the Planning Commission gave high priority to irrigation projects. Bhakra Nangal, Damodar valley, Hirakud, Rajasthan canal, Kosi, Nagarjuna Sagar, Thungabhadra, Malaprabha, Ghataprabha, etc. were the irrigation projects taken up during the first two Five-Year plans. Ramganga, Sriram Nagar, Ukai, Kadam, Tehri, Jayakwadi, Gandhi sagar, etc. were initiated later. From these multi-purpose projects, numerous canals were built up for irrigation purpose.

Although the cost of constructing canals is high, once the construction is completed, the operational cost would be less rendering them a cheap source of irrigation. Canal irrigation is important in states such as Punjab, Haryana, western Uttar Pradesh, Assam, West Bengal, Orissa, Andhra Pradesh, Telangana, Karnataka, and Tamil Nadu. The canal network includes the main canals, branch canals, major distributaries, minor distributaries, canal outlets, water courses, and field channels.

Lakes

A lake is an inland water body with still water of considerable size. Lakes can serve as a source or termination point for rivers or smaller streams. Unlike lagoons, estuaries, or backwaters, which are part of seas or oceans, lakes are unique as they are not part of any oceans. A lake is bigger and deeper than a pond.

Ponds

A pond is also a body of still water either natural or human-made but smaller than a lake. Ponds often carry shallow water with aquatic flora and fauna.

Tanks

Tanks are excavated storage structures for storing water. They form an important source of water in many Indian villages. In South India, tank irrigation is highly developed and popular. Because the rivers of South India are seasonal, the construction of perennial canals may not be feasible. Moreover, digging of canals may be difficult and expensive on the South Indian rocky and undulating terrains. The hard rocks of the region do not allow water from tanks to seep through, and hence, tank irrigation has been successful in South India.

1.8. GROUNDWATER RESOURCES

Groundwater is an important life supporting resource. It is estimated that groundwater accounts for more than 95 per cent of all the fresh water available for use. In India, nearly 95 per cent of rural people rely on groundwater for their drinking water requirements. For about half of irrigated croplands, ground water is the source of water. Ground water is the source of water for approximately one third of industrial water use. In India, water flows through most rivers during summer (barring the Himalayan rivers) depend upon groundwater. Rivers obtain ground water as springs, which flows to them as base flows.

In 1997, the Central Ground Water Board of India in collaboration with the respective State Departments assessed the ground water resources of India at 43.0 M ha-m (Chadha, 1999). The availability of renewable ground water resource for irrigation has been computed at 36.09 M ha-m. Although groundwater utilization played a major role in expanding irrigation in the 1980s, its uncontrolled exploitation has led to serious depletion of water tables in many parts of the country.

Nature of Groundwater

The pore spaces present in the soil, gravel, and rocks are filled with air and water. Water can saturate the tiny spaces between the soil materials or the crevices or fractures in the rocks. Water that percolates into the ground passes through a *zone of aeration*, also called *unsaturated* or *vadose zone*, where the open spaces between soil particles are filled with both air and water. This is the zone just above the water table. Below the zone of aeration, a *zone of saturation* could be seen where all the soil pores are filled with water. The upper surface of this saturated zone, which is subjected to atmospheric pressure is known as the *water table* or

phreatic surface. This level may be just below the ground level or many metres deep below the ground level.

Groundwater moves through the subsurface supplying water to streams and lakes. In the zone of aeration, *capillary water* that moves upward from the water table by capillary action occurs. When we say *soil moisture*, we mean water in the ground, but above the water table in the soil layers. This water can move slowly in any direction from a wet particle to a dry one. While most plants rely on moisture from precipitation that is present in the unsaturated zone in the soil, their roots may also tap capillary contribution from the underlying saturated zone. This is a common feature where water table is high.

The water occurring below the ground water table in saturated condition is commonly called *ground water* and is available for pumping and extraction by other means. It is stored in the interstices of the soil or rocks. It should be borne in mind that ground water reserves are not in the form of lakes or streams of water inside the ground. It is similar to water stored in a sponge, not visible but that can be extracted. Most groundwater occurs as aquifers. *Aquifers* are underground layers of porous soil or rocks saturated with water from above or from structures sloping towards it, which yield significant amounts of water.

Confined and Unconfined Aquifers

An aquifer, the underground formation of porous soil or rocks that are saturated with water, may be confined or unconfined (Fig.1.5). *Confined aquifers* occur where the groundwater system is between layers of clay, dense rock, or other materials with very low permeability. This impermeable layer shows very low intrinsic permeability and acts as a layer confining the underground aquifer. Any such impermeable water bearing soil or rock strata is called an *aquiclude*. The geologic strata such as silts and mudstones that are slowly permeable and retard groundwater are called *aquitards*.

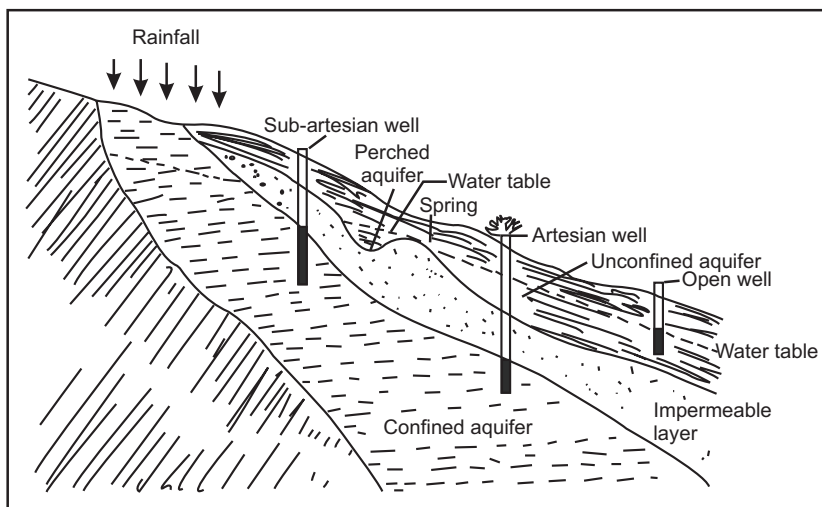


Fig.1.5: Aquifers, wells, and recharge areas.

Ground water occurring in confined aquifers may be very old, accumulated long ago, probably thousands of years ago. Ground water within a confined aquifer occurs under pressure greater than atmospheric pressure. When such a confined aquifer is pierced through a well, water rises in the well due to release of pressure within the confined aquifer. The level up to which water rises in the wells is known as *potentiometric* level. This potentiometric level indicates the magnitude of pressure within the confined aquifer. If the potentiometric level is above the ground water surface, a flowing well known popularly as *Artesian well* results (A flowing well of this kind was first observed in Artois in France, hence the name Artesian well). Such confined aquifers are known as *Artesian aquifers* and the confined pressure is also known as *Artesian pressure*.

An *unconfined aquifer* is not confined and does not have a low-permeable structure above it. Water in unconfined aquifers may have arrived recently by percolation. Note that the top layer of an unconfined aquifer is called the 'water table'. What we observe as the water level in a large dug well is nothing but the water table.

Perched Aquifer

Perched aquifer is a special case of unconfined aquifer. A perched aquifer is formed when the infiltrated rainwater is intercepted within the zone of aeration by an impermeable layer close to the surface (20cm to 100cm) and a local zone of saturation is formed. It usually covers a limited area. The upper surface of such local zone of ground water is known as *perched water table*. An impermeable layer separates the perched groundwater layer from the more deeply located groundwater table. Such aquifers usually occur at higher elevations. Although the perched aquifer exhibits all the characteristics of unconfined aquifer, its extent is limited to the dimensions of impermeable layer, and therefore, its storage is also limited. When attempting irrigation in soils with impermeable layers not far below the root zone, caution is required not to apply excess water as the perched water table may rise rapidly causing troubles, sometimes, even leading to landslides.

Depth of Groundwater Table

The depth of groundwater table varies greatly from place to place, mainly due to changes in topography of the area. In the same place, the depth may vary in time. Immediately following a heavy rainfall or irrigation, the groundwater table rises. It may even reach and saturate the root zone of crops, which if prolonged, is disastrous. With respect to the depth of ground water table, two extreme situations can be visualized. In certain situations, the groundwater table appears at the surface as in swampy areas. This is a case of *open groundwater table*. At the same time, the groundwater table can also be very deep and distant from the root zone, as happens after a prolonged dry period.

The Groundwater and Surface Water Connection

Groundwater and surface water are interconnected. In fact, it is often difficult to separate the two as they nourish each other. If we understand the hydrologic cycle,

we could better understand the connection. Water entering the soil can percolate deeper to reach groundwater, which can discharge to surface water or return to the surface through springs, seeps, wells, and marshes.

Groundwater that comes out from unconfined shallow water table aquifers is the major source of drinking and irrigation water. Unconfined aquifers also interact closely with streams, sometimes flowing into a stream or lake, and sometimes receiving water from the stream or lake. An unconfined aquifer that feeds streams is actually providing the stream's *base flow*. As the stream gains water, it is called a *gaining stream*. Ground water recharge and utilization are important. Ground water recharge occurs from rainfall, when it is received on the surface and the water infiltrates and percolates downward. It is estimated that 5-50 percent of annual precipitation percolates down and recharge groundwater. This much variation (5-50%) in the quantity of recharge is due to the influence of climate, land use, soil type, geology, and many other factors on actual recharge. In some areas, streams and rivers recharge aquifers through streambed infiltration, thereby losing considerable amount of flow. These types of streams that lose a part of flow while recharging ground water are called *losing streams*. In normal situations, discharging and recharging of ground water occurs depending on hydrologic conditions, and a balance is reached naturally.

1.9. GROUND WATER EXTRACTION

Groundwater is utilized in many ways. A common way of extraction of ground water is through wells and tanks and by installing water lifts and pumps. Several other methods are also adopted in different parts of the world.

Wells

A *water-well* is a hole made into the water bearing formation below the ground surface to extract ground water for domestic or irrigation purposes. Based on the type of well made, water from the surrounding confined or unconfined aquifer flows into the well. Water is drawn out of the well by a motor driven pump, a pump driven by a wind mill, or a manually operated lift pump. Compared to *domestic wells*, *irrigation wells* hold large volumes of water.

When water is pumped out, water level in the well drops down, and flow from the surroundings continues until the pressure difference becomes zero. The level of water in the well before pumping is equal to the elevation of the water table (or *piezometric level*), and this level is called the *static water level*. Pumping water from a well lowers the water table around the well and creates a *cone of depression* (Fig. 1.5). The cone of depression is small and shallow for wells from productive aquifers. An irrigation well pumped for long hours withdraws so much water that the cone of depression extends to vast stretches of areas. The level of water in the well after pumping is called the *pumping water level*. This level is variable and changes with pumping. The vertical difference between the static water level and pumping water level is called *drawdown*. The quantity of water that can be drawn from a well is called its *yield*. The drawdown affects the yield of a well.

The distance from the centre of the well within which the original water table is lowered due to pumping is called the *radius of influence*.

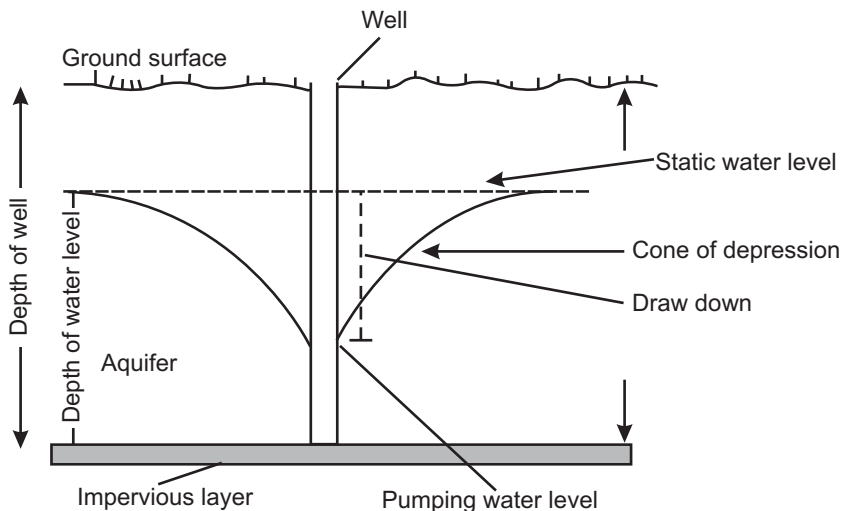


Fig. 1.6: Cone of depression- hydraulics of water flow into an open well.

If a number of wells are located closely, *interference* can occur, causing more lowering of a water table than that from widely spaced wells. Interference can draw water levels so low that pumping costs would be greatly increased. There are several kinds of wells—dug wells, driven wells, drilled wells, bore wells, and filter point wells.

Dug Wells: Dug wells or open wells have been in use from time immemorial for drinking as well as irrigation purposes. These wells are dug down to the water bearing strata to extract ground water. Dug wells mainly draw water from unconfined aquifers. The shape is circular or rectangular. The diameters of circular wells range from 1.5m to 6m, and the depth from a few meters to about 6m. Because of the large diameters of the dug wells, they can yield relatively large quantities of water from shallow sources. Water percolates into the wells from all sides and the bottom. The earthen sides of the wells should be protected from caving in by circular cement rings or burnt earthen rings. In lateritic soils, no such circular lining is given.

Driven Wells: A driven well is constructed by driving a pipe and well point through the soil into an aquifer. The *well-point* is the lower end of the pipe to which a screen is attached, which consists of a length of pipe with many perforations that allow water to enter the pipe but exclude soil material. Driven wells are circular in shape with diameters ranging from 3cm to 10cm, and are usually shallow. Construction of driven wells is recommended in areas where the water table is shallow and where the overlying soil is not hard so that a pipe can be driven easily without damaging the well point. Often, reciprocating suction type pump sets are used to draw water from these wells. As the yields from driven wells are low, their use is limited to domestic water supplies.

Drilled Wells: Drilled wells, popularly called *tube wells*, are made by using drilling machines. These are deep wells extending up to 100m or more with a diameter ranging from 15 cm to 45cm. Tube wells may penetrate more than one aquifer. When the drilling goes to the deeper layers of greater water bearing capacity under pressure, a true artesian effect is created causing the water to rise in the well to a considerable height. This increases the capacity of well. They are lined with metal casing pipes. The well casing in the aquifer is perforated and water from the aquifer flows into the well through the perforations in the pipes.

Bore Wells: Ordinary method of construction through drilling is not possible for tube wells in hard rock areas. Tube wells made in hard rock areas with special drilling mechanism are called **bore wells**. Bore wells are also circular usually with 15cm diameter and normally drilled down to a depth of 60m or more. Normally, these tube wells are not cased as the hole is made through hard rocks except perhaps in the top weathered portions. Bore wells tap water from saturated joints, fissures, and fractures in hard rock region. Bore wells usually yield 3-10 litres per second, and for water extraction, submersible pumps are generally used.

Filter Point Wells: Filter point is a shallow tube well sunk in alluvial areas to depths varying from 10m to 15m, with a diameter of 10cm to 15cm. The bottom portion of the bore is fitted with a strainer to tap water from the aquifer. Filter points are possible only at locations where the pumping level is not lower than 7m below the ground level or the level of the pump. They are normally sunk in sandy aquifers and are provided with strainers. River basins with sandy formations and high-water table are ideal for sinking filter point wells. Filter points normally used are metal, coir, and slotted pipes.

Springs

A spring is a point on earth's surface where groundwater from an aquifer discharges naturally as a stream like flow. Springs are usually found in mountainous regions at the toe of hill slopes, along depressions such as stream channels, and when the ground surface intercepts an aquifer. In most cases, springs form the headwaters of streams.

Dependent upon the permanency of water flow, springs may be ephemeral (intermittent) or perennial (continuous). In most cases, the discharge is sufficient to flow as a small brook or streamlet. In a watershed, springs can provide local sources of water for upland inhabitants, and are useful indicators of the extent of aquifers. In many remote and mountain regions, springs are the safest and most dependable source of water for domestic use. However, these springs should be dependable and continuous in that they should flow throughout the year with considerable quantity of water. Note that springs from many perched aquifers found on hill slopes are of not much use as they flow only during the rainy season. There are several kinds of springs. A common classification of springs is given below.

Depression Spring: Due to depressions on the slopes, ground water emerges out through some openings and flow as a small streamlet.

Contact Spring: In this case, rocks may be having only a mild slope. In some cases, rocks may be porous, for example, sand stone. Water comes out as a sheet of water. Water comes out at the contact point between the rocks, and hence the name contact spring.

Artesian Spring: The hard rocks may have fractures in a series. The fractures may be far off. Rain enters through these fractures. Due to elevation differences, it comes out under pressure.

Seeps: When the flow is very mild and not visible, the resulting wet areas are called *seeps*. The paddy fields in the valleys will be moist throughout the year as the ground water flows mildly or seeps.

Horizontal Wells

A horizontal well can be considered as a cased spring. Horizontal wells are drilled at potential sites where springs, seeps, or traces of water are found. A horizontal rig boring is used to drill a hole and install a steel pipe casing into a mountain or hillside to tap a trapped water supply. Presence of *phreatophytes* (plants that penetrate the groundwater reserves at great depth), dried up springs, and favorable geology are indicators used to select the drilling sites. Horizontal wells tap the aquifer with precision and safety without pumping. Maintenance costs and other problems are also less. If the flow is very low, a storage tank can be constructed to accumulate water.

Qanats: Qanat is a horizontal tunnel dug to tap underground water, especially from an alluvial formation from a mountainous region. The qanat system has been in use in various arid and semi-arid regions of the world. These horizontal wells bring water to the surface without any pumps. They vary in length depending on the depth of the aquifer and the slope of the ground. A qanat is composed of three important parts: one or more vertical head wells dug into the water bearing layers to collect water, a gently downward sloping underground horizontal tunnel leading the water from the head wells to a lower point at the surface, and a series of vertical shafts between the ground surface and the tunnel for ventilation and removal of excavated debris. The length of qanat from the head well to the opening is usually in the range of 1 km to 4 km. It is believed that about 3000 years ago, the Persians learned to dig 'qanats' (probably, the forerunner of the word 'canal') to bring mountain ground water to arid plains.

Surangam: Surangam is a shaft or channel dug horizontally through a laterite hillock from which water comes out as a flow, which is collected in a storage tank. Construction of surangam is common in the northern parts of Western Ghats. It is similar to a qanat, but much less in length; most surangams are of 8m to 10 m length only and maximum recorded length is about 300m. The head water wells and shafts of qanats are also absent. The location of surangams is decided based on the topography and soil characteristics. Surangams are constructed based on local expertise. Sometimes, many subsidiary surangams are excavated inside the main surangam. Construction is done in such a way that there will be a slight slope

towards the opening of the surangam for the smooth flow of water. Water coming out of the suranagm is collected in a small storage tank constructed near the outlet of the surangam from which water is guided to lower reaches for various purposes.

1.10. CONJUNCTIVE USE OF WATER

Conjunctive use of multiple sources of water needs to be considered to optimize the development, management, and conservation of water resources within a watershed or farm. Conjunctive use of water means harmoniously combining the use of water from multiple sources, for example, surface and ground water, in order to minimize undesirable physical, environmental, and economical effects of each source and to optimize the water demand-supply balance. Common conjunctive use systems are:

- ◆ Surface water and groundwater
- ◆ Rainfall and irrigation water
- ◆ Saline water and freshwater
- ◆ Fresh water and used water

Surface Water and Groundwater System

There is immense scope for conjunctive use of surface water and ground water resources. Considerable quantities of water diverted for irrigation from reservoirs are lost through seepage and percolation from canal systems. These 'lost water' and return flows from irrigation contribute to ground water recharge. The water thus stored in ground water reserves can be extracted and conjunctively used for irrigation in command areas to tide over the peak demands and water shortages. Use of ground water helps to reduce peak demands of irrigation, and therefore, size of canals can be reduced leading to savings in construction costs. Supplemental groundwater use promotes multiple cropping ensuring proper irrigation scheduling. It also helps to go for early sowing of crops, even if rainfall is delayed. Similarly, increased water resources ensure water supply to tail end areas and higher elevation areas. In waterlogged areas, groundwater exploitation lowers the water table and reduces the danger of water logging.

Conjunctive use of surface and ground water when integrated with an artificial recharge project reduces the need for lining canals as seepage from canal recharges ground water. Moreover, during periods of peak water demand, as irrigation requirement can be met from surface water sources, pumping is not needed and power saved can be used for other purposes.

There are several problems in the development of surface water resources compared to ground water resources. Soil erosion in the catchment areas causes siltation in surface water reservoirs, and consequently, their storage capacity is reduced. When the vegetative cover is lost, probability of sediment accumulation is greater in surface reservoirs. Another problem is exposure of large open areas of water to evaporation leading to heavy losses of water. Distribution of water from the reservoir requires the construction of a canal system, which add expenses

to the total cost. Flooding and destruction of vast stretches of forests and good agricultural lands are often highlighted. Rehabilitation of displaced people is another great worry for the Government.

Ground water development is suggested as a partial solution to the problems of surface water development. As we know, groundwater is not exposed, and therefore, not prone to evaporation. Unlike surface storage, siltation and consequent reduction of storage capacity will not be a problem. It is considered less harmful to the environment, and in most cases, allow the users to have a natural water distribution system.

Nevertheless, this alternative is often not considered well. In practice, surface water storage is always preferred to groundwater development, and many reasons can be attributed for this preference. Large water demand is usually met from big irrigation schemes supplied from surface water reservoirs and groundwater aquifers seldom offer large storage capacity to absorb big volumes of flow in a short period. Another major reason is political. As Pallas (1995) remarked, big surface storage dams or reservoirs require large investments and better political visibility; therefore, it is often preferred, and it offers much opportunities for private profit and corruption.

Conjunctive use of Rainfall and Irrigation

In most parts of the world, rainfall occurs during a particular season of the year only, and during this time, water requirement of plants is met from rainfall. Because of seasonal rainfall, peak runoff corresponding to substantial part of the total discharge of rivers also occurs during this season. Although water availability is high during the rainy season, total water demand may be low. Therefore, the efforts must be to transfer excess water from the high supply season to the high demand season. This can be achieved by storing surface water in ponds, tanks, and reservoirs. Direct utilization of rainfall, storage of excess rainfall, and irrigation during rainless periods using the saved water are excellent examples of conjunctive use of rainfall and irrigation.

Conjunctive use of Freshwater and Saline Water

Sometimes, we may confront situations where good quality water is not available for irrigation in adequate quantities but saline water is available in excessive quantities. In such situations, conjunctive use of freshwater and saline water is a good option. Many choices are available like mixing of saline water with good quality water to obtain irrigation water of medium salinity, cyclical use of fresh water and saline water, use of good quality water at the more critical stages of growth, particularly during germination and seedling establishment, and use of saline water at the stages where the crop has relatively more tolerance. For example, certain crops like pigeon pea, sorghum, and mustard tolerate saline water well once the good quality water is used for pre-sowing irrigation to leach out the salts for helping seed germination.

Conjunctive use of Fresh Water and used Water

Any domestic wastewater produced, for example, water from bathroom sinks, showers, tubs, and washing machines excluding sewage is grey water. Probably, the term “wastewater” is a misnomer as grey water can be reused, and a more appropriate term for waste water would be “used water”. By appropriate treatments, grey water including water from laundry and toilet flushing can be reused. There is no harm in using treated grey water for irrigating both food and non-food crops. The elements such as nitrogen and phosphorus in grey water are good sources of nutrients for these plants. Sometimes, fresh water and treated used water can be mixed and reused. Reusing grey water reduces the need for fresh water, reduces water footprint, and optimizes domestic as well as industrial requirements and costs.

1.11. INTEGRATED WATER RESOURCES MANAGEMENT

Water has a basic function in maintaining the integrity of the natural environment. However, in the modern world, the problems to maintain the integrity of water are many. Population growth, urbanization, industrialization, expansion of agriculture, tourism, and climate change—all are putting water under great stress. The pressure on water resources highlights the hydrological, social, economic, and ecological inter-dependencies in rivers, streams, lakes, ponds, and aquifers. To address the multi-faceted nature of water management, many countries are now introducing an integrated approach to manage water resources at the national and watershed level, which includes, among other things, improved institutional arrangements and working practices. The problem of diminishing water supplies and ever-increasing demands must be confronted and tackled realistically. The traditional reductionist approach of water resources management is no longer viable, and a more holistic approach is essential. This is the background of the Integrated Water Resources Management (IWRM) approach, which has now been accepted internationally as the way forward for efficient, equitable, and sustainable development and management of the limited water resources of the world. Rahaman and Varis (2005) discussed the evolution of the concept of Integrated Water Resources Management (IWRM) over the past three decades and the prospects of IWRM in resolving the current water crisis.

According to the Global Water Partnership (GWP), a network of organisations working in the field of water resources management including government institutions, agencies of the United Nations Organization, professional associations, research institutions, non-governmental organizations, and private agencies, Integrated Water Resources Management (IWRM) is “a process that promotes the coordinated development and management of water, land, and related resources, in order to maximize the resultant economic and social welfare in an equitable manner without compromising the sustainability of vital ecosystems” (GWP, 2009). At the river, stream, and aquifer level, IWRM can be defined as “a process that enables the co-ordinated management of water, land, and related resources within the limits of a watershed so as to optimize and equitably share the resulting

socioeconomic well-being without compromising the long-term health of vital ecosystems”.

Multiple uses and Needs of Water

Water is of fundamental importance for food production, drinking, sanitation, and hygiene. Agriculture is the major user of water worldwide, about 69 percent of total withdrawals! Domestic and municipal use amounts to about 10 percent and industry uses some 21 percent. Water use varies considerably around the world. However, substantial amount of water must be present in the environment for healthy ecosystems. An IWRM approach focuses on three basic pillars that together act as the overall framework:

- ◆ *Social equity*: To ensure equal access for all users, especially marginalized and poorer user groups, to an adequate quantity of good quality water necessary to sustain human well being.
- ◆ *Economic efficiency*: To bring maximum benefit to the highest number of users possible with the available financial and water resources.
- ◆ *Ecological sustainability*: Aquatic ecosystems are acknowledged as users and that adequate allocation is made to sustain their natural functioning.

Water for social and economic development is clearly linked to the IWRM focus on social equity, economic efficiency, and ecological sustainability. Water for social development includes the provision for education and health care. Without clean water and good sanitation facilities in educational institutions and hospitals, social development is blocked. Water is also important for economic development, especially for the generation of energy. It is needed for many forms of energy, particularly hydropower. It is also used extensively for cooling thermal and nuclear power stations. Energy is also needed for pumping, including extraction of water from underground aquifers. Industry needs water in a big way. The industries, through pollution and abstraction, have an influence on water quality that affects both downstream users and natural ecosystems. All these have significant implications for water resources management.

Interlinking of Rivers

The concept of linking of rivers or inter-basin transfer of water has been accepted as a partial solution to address water availability in water scarce regions. It is essentially based on the availability of surplus water in a donor river especially at the point of diversion to a deficit river basin. For the sustainable development of a region confronting water deficits, interlinking of rivers is an effective management strategy for surface water resources. It is an economically viable, technically feasible, and environmentally sound option. The surplus or deficit in a basin is determined based on availability at 75 per cent dependability, especially for import, export, and existing and future needs. A river basin is said to be reasonably in surplus, if the surplus water is available after meeting the irrigation needs of at least 60 per cent of the cultivable area in the basin. Only this much water from a basin can be diverted to deficit basins.

In India, the National Water Development Authority (NWDA) indicated that Himalayan rivers, especially, the Brahmaputra and the Ganges have remarkably surplus water, and therefore, proposed transfer of water from these surplus basins to deficit basins in the peninsular region.

The proposed inter-linking of rivers (ILR) project in India has plans to transfer water from surplus to water-deficit areas to get rid of droughts and floods, and to utilize water which otherwise would go as waste into the sea. It is an inter-basin water transfer project with plans to connect 37 rivers of 20 major basins across the nation through 31 links and canals having a total length of 9,000 km. The scheme involves two components, a Himalayan component with 14 links and a Peninsular component with 16 links. However, the issues of ecological concerns and massive displacement of people must be properly addressed before embarking on such mammoth projects.

1.12. THE VIRTUAL WATER AND THE WATER FOOTPRINT

For understanding and managing global freshwater reserves effectively, it is important to understand the hidden water use behind every consumer products and services. The recently emerged concepts of *virtual water* and *water footprint* are intended for this purpose, and these are now accepted as international development tools to quantify the effects of freshwater resources on global production, consumption, and trade of various food and non-food commodities.

The Virtual Water

Humans utilize water directly for drinking, cooking, bathing, washing, and for toilet uses but indirectly much more for producing food and non-food commodities. Water is spent in a big way for the production of agricultural commodities, processed food products, paper, cotton, rayon, and innumerable industrial products. Traditionally, the stakeholders of water resources considered water from the production side only, especially by quantifying the three major areas of water use— agriculture, industry, and domestic sectors. Of late, however, it has been recognized that it is a limited way of looking at global water use because agricultural products and manufactured goods are not always consumed in their country of origin. In other words, international trade of agricultural and industrial products created the flow of what is known as ‘virtual water’ or ‘embedded water’ from one country to other countries.

The concept of “virtual water” is increasingly used now to highlight the way in which water scarce regions manage to reduce the water demand. It was originally developed considering the water scarce countries of the Middle East and North Africa, who import a large portion of their food. John Anthony Allan coined the term virtual water in a seminar at the University of London in 1993 for what was earlier described as ‘embedded water’ (Allan, 1993; 1996). The amount of water used in the production process of a commodity or service delivery is called *virtual water*. The water is virtual because it is no longer present in the product. In other words, virtual water is the water ‘hidden’ in the products, services, and processes people buy and use every day. Although water has been consumed throughout the

value chain for making a product or service, often, the end-user of such a product or service may not see or feel the water utilized for their making. Thus, the water utilized becomes virtual! By importing food, water scarce countries could reduce substantially the water demand in domestic food production and compensate for the scarcity of water. In other words, for them, importing food is almost equivalent to trading water!

Blue, Green, Grey, and Black Water: Virtual water can be further divided into ‘blue’ water, ‘green’ water, ‘grey’ water, and ‘black’ water. Blue water and green water are natural. Blue water exists in streams and rivers as flowing water and as still water in storages of freshwater existing at the land surface as lakes, ponds, tanks, and aquifers. Blue water can be utilized for irrigating crops or for direct domestic and industrial uses. Please note that ‘land’ includes green water as it is a part of it, but does not include blue water or irrigation supplies. The demands for blue water are diverse, because it can be consumed in all economic activities including the generation of energy. It is normally over-used and vulnerable to over-consumption.

Green water means the precipitation and water seen on and in soil, which are directly relevant to local vegetation and crops. It is the so-called ‘effective rainfall’ stored in the soil. A part of this green water is returned to the atmosphere through evapotranspiration. Green water cannot be pumped or moved except as virtual water embedded in food supply chains. The crops access green water for growth and development and transport it as virtual water to food consumers. Green water consumption is limited in the sense that it is utilized for sustaining natural vegetation and producing local crops. For the poor farmers of rainfed area, green water is the only source for sustaining their crops. However, the problem is growers can use the green water actually available for the current season only. Unfortunately, the costs for mobilizing or protecting green water are not taken into account in our food systems. Nevertheless, the important thing we must know is that a unit of green water can produce as much food as a unit of blue water!

Water consumption is of two types—food-water consumption and non-food-water consumption. Water consumed for food production can be blue or green. However, non-food production always uses blue water and sometimes recycled blue water. Re-use of green water is seldom possible, but reuse of used blue water (note that once blue water is polluted, it become grey water!) is often attempted. In this respect too, high rates of blue water re-use can be achieved in non-food systems than food system. In water deficient countries such as Israel, blue water reuse of as high as 80 percent have been achieved in non-food-water uses.

Table 1.4: Average virtual water content of some selected products (L/kg)*

<i>Products</i>	<i>India</i>	<i>World average</i>
Wheat	1654	1334
Rice (paddy)	2850	2291
Rice (husked)	3702	2975
Maize	1937	909

<i>Products</i>	<i>India</i>	<i>World average</i>
Barley	1966	1388
Soybean	4124	1789
sugarcane	159	175
Cotton lint	18,694	8242
Coconut	2255	2545
Coffee (roasted)	14,500	20,682
Tea (made)	7002	9205
Beef	16,482	15,497
Chicken	7736	3918
Pork	4397	4856
Egg	7531	3340
Milk	1369	990
Leather (bovine)	17,710	16,656

**Selected from Hoekstra and Chapagain (2007)*

Any domestic wastewater produced, for example, water from kitchen sinks, bathroom sinks, showers, tubs, and washing machines excluding sewage is called *grey water*. The polluted water from the industrial establishments is also grey water. Do not confuse grey water with *black water*. The main difference between grey water and black water or sewage is the organic loading. Sewage has much larger organic loading compared to grey water. Probably, the term “wastewater” is a misnomer as grey water can be reused, and a more appropriate term would be “used water”. By appropriate treatments, grey water including water from laundry and toilet flushing can be reused. There is no harm in using treated grey water for irrigating both food and non-food crops. Sometimes, fresh water and treated grey water can be mixed and reused. Thus, reusing grey water helps to reduce the water foot print.

Considering the virtual water, it makes obvious sense for water-scarce countries to import basic foods such as cereals from water-surplus areas and use their own limited water resources to grow high value crops for export—such as cut flowers, strawberries, and other fruits. The foreign exchange thus earned can then be used to buy cereals through imports. In other words, importing food is equivalent to importing water in a condensed form.

Plants take up huge quantities of water but most of it is lost through evaporation and transpiration. It is estimated that to produce cereal grains, 1000–3000 m³ of water is needed per tonne of grains (1000–3000 litres of water to grow 1 kg of grains). In a similar way, the amount of water needed for producing various products can be estimated. The virtual water content of some common products is presented in Table 1.4. Note that the values are indicative and they may differ depending on the way the products are made in different countries.

We can also estimate the virtual water content of our daily diet! We may be directly drinking only 2-3 litres of water daily. However, it is estimated that to produce enough food to satisfy a persons' daily diet, 2000-3000 litres of water must be spent!

Although self-sufficiency in food grains is a dream of many nations, the current trend is away from food self-sufficiency to partial reliance of food imports. One of the main driving forces to this change is water scarcity caused by rapidly growing populations, which have reduced per capita land and water availability. At the same time, there have been increasing domestic and municipal demands on limited water resources. Some countries have also found that there are higher returns on labour in industries than agriculture. In short, it is easier and more profitable to earn foreign exchange to go for food imports rather than growing water-hungry crops. Nevertheless, it is essential that some countries of the world should produce the required food for all! Good crop and land husbandry practices can significantly reduce the amount of water needed in agriculture, both in rainfed and irrigated agriculture.

The Water Footprint

The concept of *water footprint* (WF) is of recent origin, which is closely related to virtual water. In fact, water footprint was built up on the concept of virtual water, which indicates the entire volume of water (including the virtual water) required per time unit by one or more persons, companies, or people in particular places. The global water footprint shows the total amount of water used by all human beings in a certain time unit. Most commonly, the volume of water per year is taken as a measure. Water footprint is a consumption-based indicator referring to the total amount of water including virtual water consumed by a person, group, company, or a region per unit time, mostly year.

The concept of water foot print was introduced in 2002 by Arjen Hoekstra of the United Nations Educational, Scientific, and Cultural Organization (UNESCO) as a comprehensive indicator of water use (Hoekstra, 2003; Hoekstra and Chapagain, 2007). The whole quantity of water used in the supply chain of a product has been considered. This includes actual quantity of water used for the product and the water polluted or dumped into a basin or sea. The water footprint has links with the end consumers, manufacturing companies, and traders. The concept was refined since then and from 2009, it has been promoted as a global calculation standard for water foot printing, maintained by the Water Footprint Network, an international network of governments, corporations, non-governmental organizations, and UN programmes.

Although both virtual water and water footprint can refer to the water used to produce an item, the water footprint concept can be applied more broadly. Water footprint not only refers to the volume of water used but also to which kind of water is used and to when and where that water is utilized. For example, the virtual water content of a product is the sum of the water used along the value chain. It is the freshwater 'embedded' in that product in virtual sense (not in real sense).

In other words, it is the volume of water consumed or polluted for producing the product, measured over its full production chain.

The concept of water footprint is similar to carbon footprint. Water footprint can be defined as the total volume of freshwater used to produce the goods and services consumed by individuals, households, communities, cities, businesses, or even countries. WF calculation takes into account the volumes of blue, green, and grey water that have been directly used or indirectly lost (evaporated or polluted) in the process of production of commodities and services and in the livelihoods and domestic activities of individuals and communities. Total volume of water consumed, evaporated, and polluted are accounted for estimating water foot print.

It is common to analyze and separate water foot print into three components, namely, blue water footprint, green water foot print, and grey water foot print.

Blue water footprint: The amount of surface water and groundwater used to produce an agricultural or industrial product.

Green water footprint: The amount of effective rainfall and soil moisture consumed by plants and crops in their cultivation. The green water footprint is related to the agricultural sector as only plants can utilize green water.

Grey water footprint: The amount of freshwater required to dilute the wastewater generated in the production of various items to maintain water quality within the safe limits. It can also be the volume of water polluted during the production process.

Computing water footprints has increased awareness of the concept of virtual water and its role in international trade. Water footprints help individuals and groups as they reveal their water use patterns. We also get information on the water required for the manufacture and processing of various commodities. Water contaminated during the production process is also considered. In addition, after assessing the water footprint of a product, one can ascertain whether the production process is sustainable under the existing ecological conditions.

Mekonnen and Hoekstra (2011) estimated the annual global water footprint in the period 1996–2005 as 9087 Gm³ (74% green, 11% blue, 15% grey). They also estimated the water footprint of the global average consumer during the same period as 1385 m³/yr. About 92 percent of the water footprint is related to the consumption of agricultural products, 5 percent to the consumption of industrial goods, and 4 percent to domestic water use. The largest contribution to the water footprint is from the consumption of cereals (27%), followed by meat (22%), and milk products (7%).

Based on the concept of virtual water, several countries including the USA and Brazil are able to 'export' billions of litres of virtual water each year through various farm products they grow and trade. At the same time, countries such as Japan, UAE, Saudi Arabia, and Egypt 'import' billions of litres of virtual water.

Note that the water footprint of a country (m³/year) is computed as the volume of water actually used and made unavailable for other uses through the production

of a commodity to fulfill consumption demands. This footprint consists of two parts referring to the use of domestic water resources versus the use of foreign water resources: the *internal water footprint* (the part of the footprint that falls inside the country); and the *external water footprint* (the part of the footprint from other countries in the world). Please refer to the URL of Water Footprint Network for the statistics on water footprints of different countries (<https://waterfootprint.org/en/resources/waterstat/>). It gives virtual water imports and exports, green, blue, and grey water footprints and the total water footprint by both production and consumption. The per capita annual water footprint of some selected nations (based on consumption) is given in Table 1.5.

Table 1.5: The per capita annual water footprint of selected nations (based on consumption)

<i>Countries</i>	<i>Footprint (m³/year)</i>
Australia	2315
Bangladesh	769
Brazil	2027
Canada	2333
China	1071
Egypt	1341
Germany	1426
India	1089
Israel	2303
Japan	1379
Mexico	1978
Mongolia	3775
Niger	3519
Pakistan	1331
Russian Federation	1852
Saudi Arabia	1849
South Africa	1225
Sri Lanka	1256
Sweden	1428
The Philippines	1378
United Arab Emirates	3136
United Kingdom	1258
United States of America	2842
World average	1385

*Selected from Mekonnen and Hoekstra (2011)

1.13. FOOD SECURITY AND WATER SECURITY

Food security of many nations is in peril. All households should have access to adequate food for all their members. In addition to the availability of food, people must be able to access it. This condition is called *food security*. As defined and accepted in the World Food Summit held in 1996, “food security is a situation that exists when all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food that meets their dietary needs and food preferences for an active and healthy life” (FAO, 1996). *Food insecurity* exists when people do not have adequate physical, social or economic access to food as defined above. Food security is multi-dimensional and comprises of four key dimensions or “pillars” of food supplies: *availability*, *stability*, *access*, and *utilization*.

Availability relates to the availability of sufficient food, that is, the ability of the prevailing agricultural system to meet the food demand. If adequate amounts of food are produced and are ready at people’s disposal, we can say that availability is ensured. Stock levels and net trade also influences availability.

Access refers to the ability of a person to produce one’s own food or buy it, which implies having the purchasing power to do so. Access is ensured, when all households and all individuals within those households have adequate resources to obtain appropriate foods for a nutritious diet. This can be ensured through production, purchase, or donation.

Utilization means how the human body makes use of various nutrients in the food. It encompasses all food safety and quality aspects of nutrition; and its sub-dimensions are related to health, including the sanitary conditions across the entire food chain. This dimension determines the nutritional status of individuals, and hence we can say that food security integrates the notion of food safety as well.

Stability is related to individuals who are at high risk of temporarily or permanently losing their access to the resources needed to consume adequate food. Stability is in terms of availability, accessibility, and quality. This dimension incorporates issues of price stability and securing incomes for vulnerable populations. Adverse weather conditions, political instability, or economic factors such as unemployment and rising food prices may have an impact on food security status of individuals. Stability is ensured when the three other pillars of food security are maintained over time.

Like food security, the concept of *water security*, a condition of accessibility, reliability, and timely availability of adequate safe water to satisfy basic human needs is also important. Food security and water security are intimately linked, and access to water is a key to food security. The concept of water security considers the capacity of a population to make available clean water needed for them. The following definition of water security has been accepted by UN (UN water, 2013). “Water security is defined as the capacity of a population to safeguard sustainable access to adequate quantities of acceptable quality water for sustaining livelihoods, human well-being, and socio-economic development for

ensuring protection against water-borne pollution and water-related disasters and for preserving ecosystems in a climate of peace and political stability”.

Water is a critical constraint to achieve food security. Note that food security also depends on maximizing both the food and the number of jobs produced for ‘every drop of water’ used whether in irrigated or rainfed agriculture. Thus, in addition to ‘more crop per drop’, ‘more jobs per drop’ and ‘more value per drop’ are other considerations, which must find a place in the agenda.

Often, it is highlighted that much of the success of the Green Revolution came from improved productivity in areas where assured irrigation was provided through canals and groundwater sources. Although irrigated agriculture has played a significant role in increasing food production in the recent decades, its absolute contribution is still lower than that of rainfed agriculture. Irrigation contributes about 40 percent to the world’s food supply from about 17 percent of the total arable land in the world (approximately 250 M ha out of 1500 M ha). The remaining 60 percent food production comes from rainfed agriculture. In water-scarce tropical regions such as sub-Saharan Africa, rainfed agriculture is the norm in more than 95 percent of croplands and will remain the dominating source of food for growing populations. In India, out of 141 million ha of net cultivated area, 68 million ha is irrigated (48%) and 73 million ha (52%) is rainfed. Nearly, 60 per cent of the total production is from irrigated area, and the balance 40 per cent from rainfed areas.

In future, a major concern will be the availability of enough freshwater to satisfy the growing needs of agriculture and other water uses. Enough water must be ensured to grow the food needed by an expanding world population. Agriculture already accounts for about 69 percent of the freshwater withdrawals in the world, and this is often blamed as the main factor behind increasing global freshwater scarcity. It is feared that more regions will face water scarcity and will not be able to produce all the food they need locally. Increasing irrigation efficiency and water productivity—getting more yields per drop of water used—must thus become one of our top priorities.

For raising agricultural productivity in water scarce regions, irrigation is indispensable. Changing climate is also playing spoilsports. It is worrisome that water is already scarce in many countries. In such a scenario, where shall we find the water to grow the crops needed to feed the world? The answer lies in improving both crop productivity and water productivity. This is possible by using better seeds, drought tolerant/resistant crops, boosting soil fertility, judicious use of water, and implementing other improved agricultural techniques. Through more efficient irrigation methods and enhanced water harvesting techniques, water can be put to its maximum use.

Although groundwater utilization played a major role in expanding irrigation in the 1980s, uncontrolled exploitation of groundwater has led to serious depletion of the water table in many parts of the country. We have to learn many lessons from past mistakes, and in future, ground water development and utilization should be done in a sustainable manner.

Efficient irrigation methods are a prerequisite to utilize water to its full advantage. For example, surface irrigation methods in which water floods the fields and sprinkler irrigation that almost mimics rainfall can waste water. More efficient than these irrigation methods are localized methods, for example, drip irrigation, micro sprinkler, and hose irrigation, which put water only where it is needed. These water saving methods should get prominence where water is already scarce.